8b.Shielding Effectiveness and EMC Compliance Analysis using MATLAB

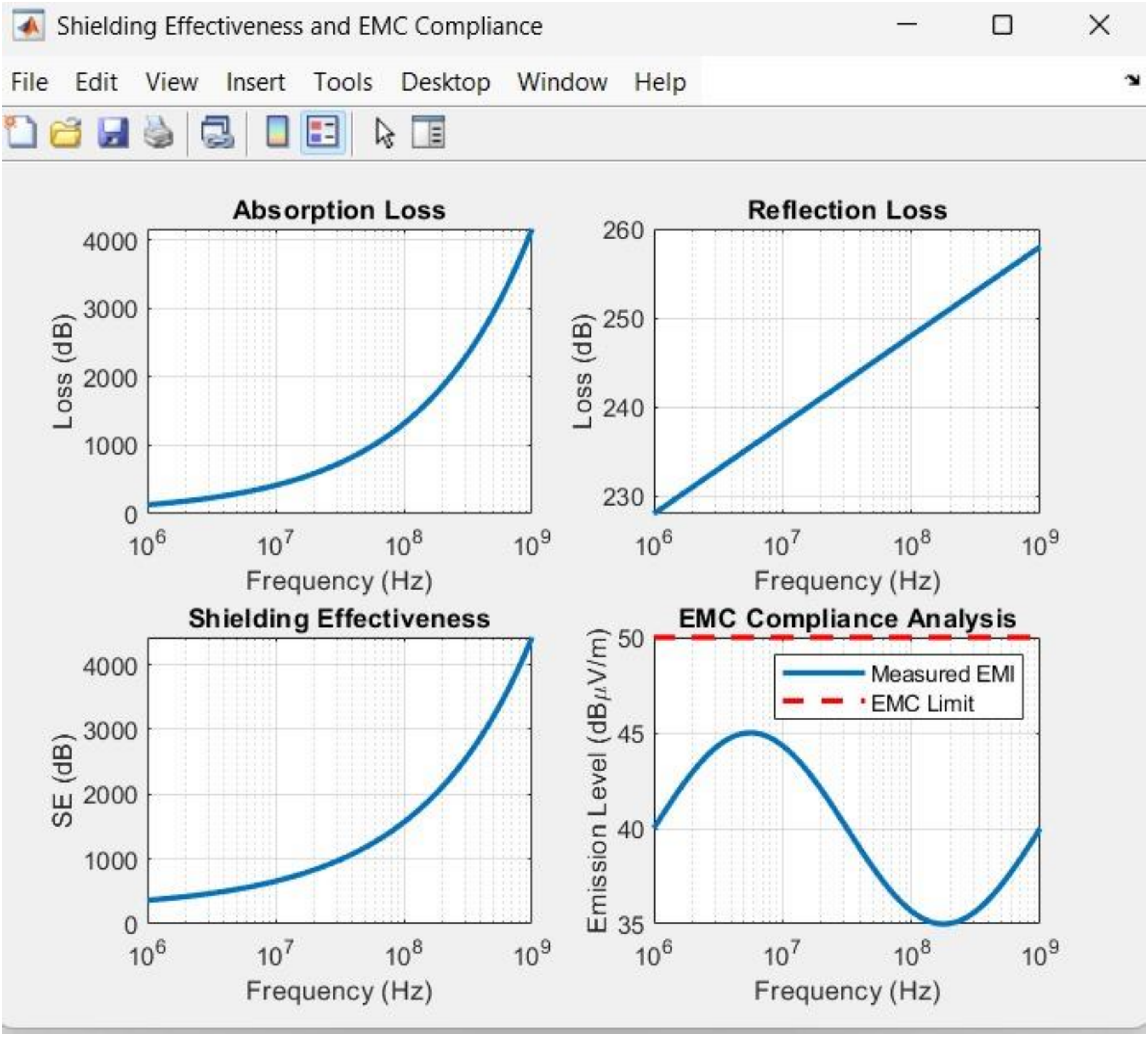
OVERALL MATLAB CODE:

```
clc;

clear;

close all;

%% Objective 1 : Shielding Effectiveness Simulation

f = logspace(6,9,1000); % 1 MHz to 1 GHz

mu0 = 4*pi*1e-7;

sigma = 5.8e7; % Copper conductivity

t = 1e-3; % Shield thickness

omega = 2*pi*f;

delta = sqrt(2./(omega*mu0*sigma));

A = 8.686*(t./delta); % Absorption Loss (dB)

%% Objective 2 : Reflection and Multiple Reflection Loss

sigma_r = 1;

mu_r = 1;

R = 168 - 10*log10((sigma_r)./(mu_r*f));

B = 5; % Approximate Multiple Reflection Loss

SE = R + A + B;

%% Objective 3 : EMC Compliance Analysis

EMC_Limit = 50*ones(size(f));

Measured_Emission = ...

40 + 5*sin(2*pi*log10(f)/3);


Margin = EMC_Limit - Measured_Emission;
```

```
Compliance = Margin > 0;
%% Display Results
fprintf('\nSHIELDING EFFECTIVENESS RESULTS\n');
fprintf('--------------------------------\n');
fprintf('Maximum SE = %.2f dB\n',max(SE));
fprintf('Minimum SE = %.2f dB\n',min(SE));
if all(Compliance)
fprintf('System is EMC COMPLIANT\n');
else
fprintf('System is NOT EMC COMPLIANT\n');
end
%% Plot Results
figure('Name',...
'Shielding Effectiveness and EMC Compliance',...
'NumberTitle','off');
subplot(2,2,1)
semilogx(f,A,'LineWidth',2)
grid on
title('Absorption Loss')
xlabel('Frequency (Hz)')
ylabel('Loss (dB)')
subplot(2,2,2)
semilogx(f,R,'LineWidth',2)
grid on
title('Reflection Loss')
xlabel('Frequency (Hz)')
```

```
ylabel('Loss (dB)')
subplot(2,2,3)
semilogx(f,SE,'LineWidth',2)
grid on
title('Shielding Effectiveness')
xlabel('Frequency (Hz)')
ylabel('SE (dB)')
subplot(2,2,4)
semilogx(f,Measured_Emission,...
'LineWidth',2)
hold on
semilogx(f,EMC_Limit,...
'r--','LineWidth',2)
grid on

legend('Measured EMI',...
'EMC Limit')
title('EMC Compliance Analysis')
xlabel('Frequency (Hz)')
ylabel('Emission Level (dBμV/m)')
```

OUTPUT:



Shielding Effectiveness and EMC Compliance
File Edit View Insert Tools Desktop Window Help
Absorption Loss
Loss (dB)
Frequency (Hz)
Reflection Loss
Loss (dB)
Frequency (Hz)
Shielding Effectiveness
SE (dB)
Frequency (Hz)
EMC Compliance Analysis
Emission Level (dBμV/m)
Measured EMI
EMC Limit
Frequency (Hz)